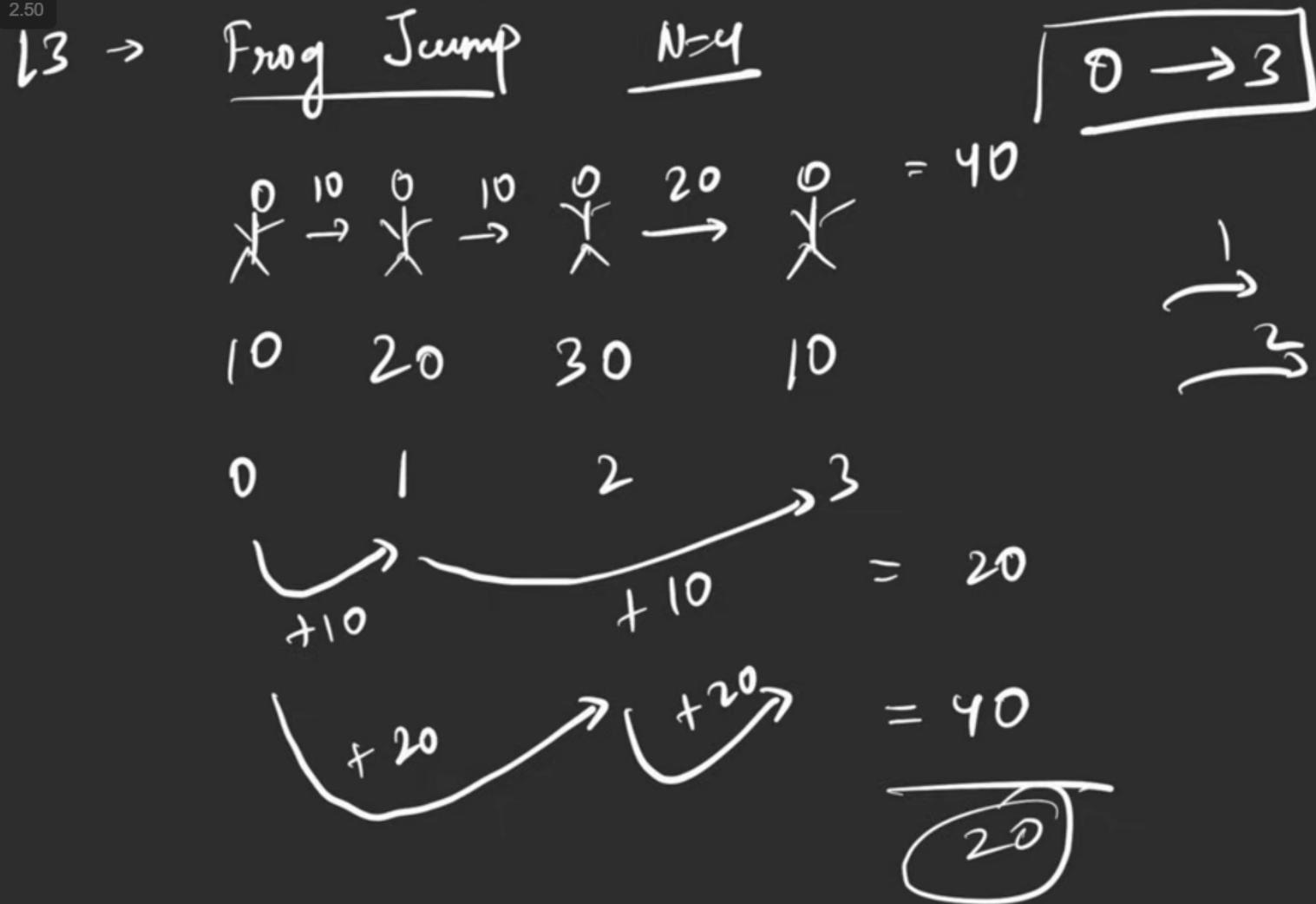
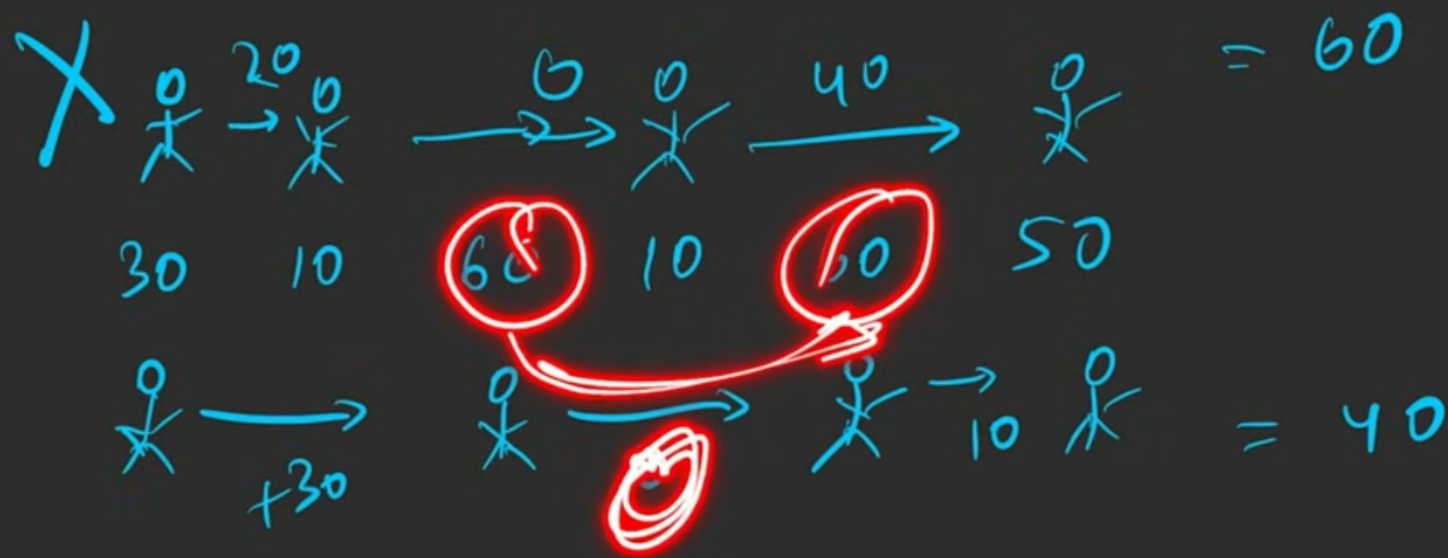


2.50



Why a greedy solution does not work $\rightarrow$?



2.50

```
f(ind)
{
    if (ind == 0) return 0;

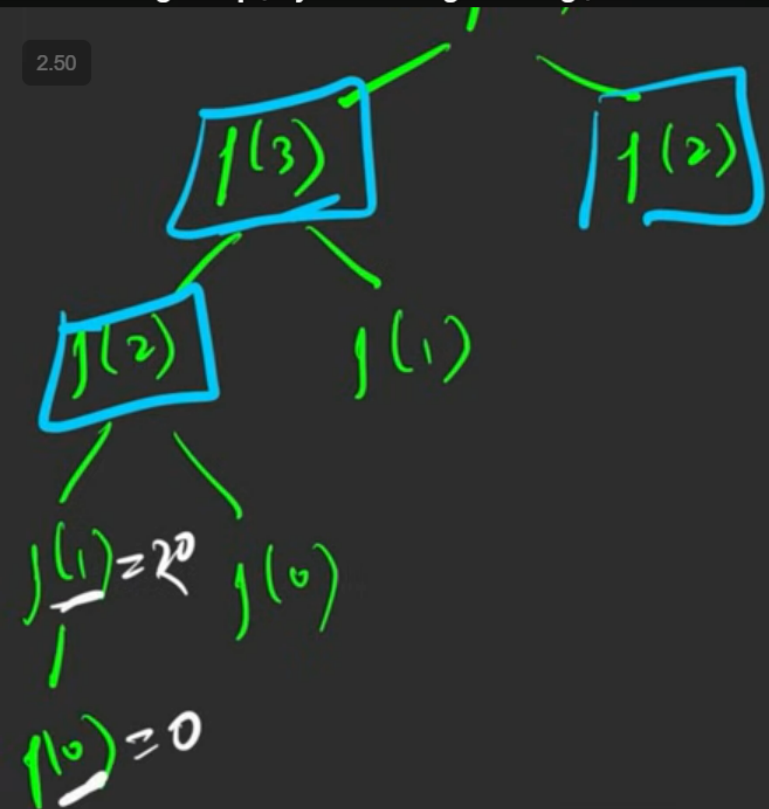
    left = f(ind-1) + abs(a[ind] - a[ind-1]);
    right = f(ind-2) + abs(a[ind] - a[ind-2]);

    return min(left, right);
}
```

(ind-1) ind



2.50



$$f(1)$$

$$\{$$

$$\text{left} = f(0) + 20$$

$$\text{right} = \text{JUMP}$$

$$\text{return } \min(20, \text{MAX})$$

$$f(5)$$

$$\{$$

$$\}$$


15:14 / 38:49

Memoization technique >



(1) ind $\rightarrow (0, 1, 2, \dots, n-1)$

(2) Do all stuffs on that ind $\begin{matrix} \xrightarrow{+1} \\ \xrightarrow{+2} \end{matrix}$

(3) Take the min (all stuffs)

$f(0) \rightarrow (0 \rightarrow 0) \rightarrow 0$ $f(n-1) \rightarrow$ min energy req to reach $(n-1)$ from 0

$f(ind)$

{

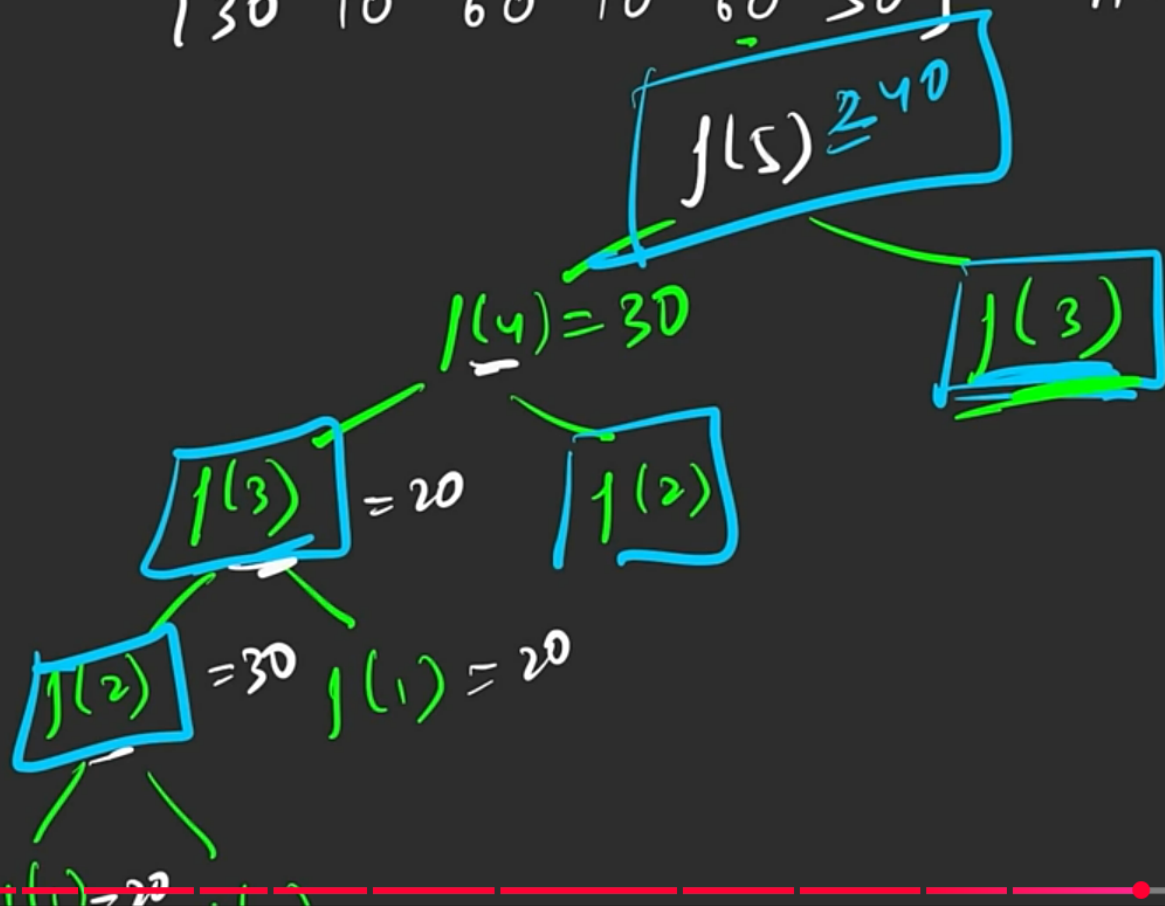
$(ind-1)$ ind

if $(ind-1) \geq 0$ return 0;



2.50

$\{30^0, 10^1, 60^2, 10^3, 60^4, 50^5\}$ $n=6$

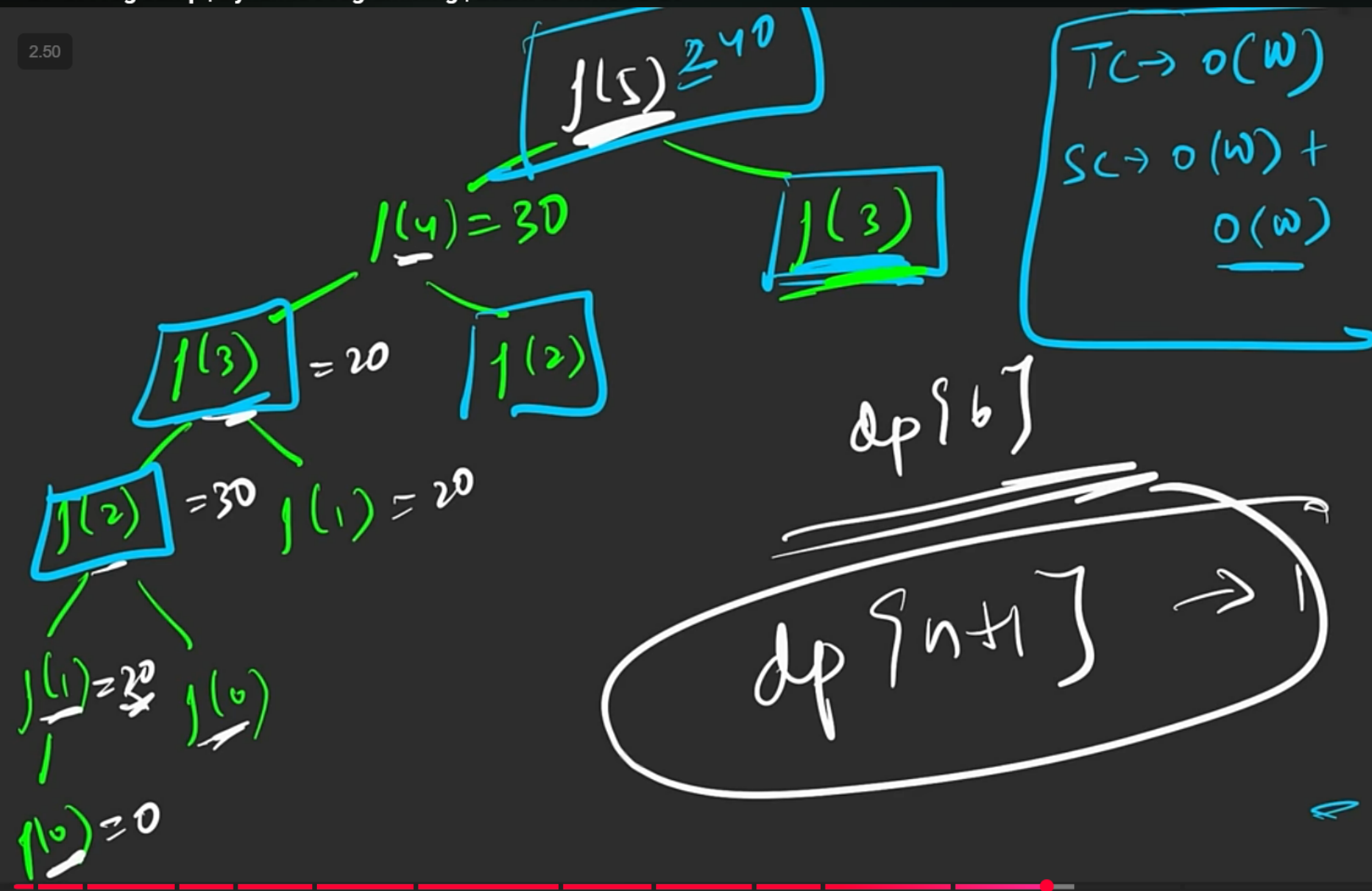


2.50

$f(ind)$
 $\{$
 $if (ind == 0) \text{ return } 0;$
 $if (dp[ind] != -1) \text{ return } dp[ind];$ (3)
 $left = f(ind-1) + \text{abs}(a[ind] - a[ind-1]);$
 $\Rightarrow \text{right} = f(ind-2) + \text{abs}(a[ind] - a[ind-2]);$
 $return dp[ind] = \min(left, right);$ (2)



2.50





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20

0

Explanation Of Sample Input 1:

For the first test case,
The frog can jump from 1st stair to 2nd stair ($|20-10| = 10$ energy lost).
Then a jump from the 2nd stair to the last stair ($|10-20| = 10$ energy lost).
So, the total energy lost is 20 which is the minimum.
Hence, the answer is 20.

For the second test case:
The frog can jump from 1st stair to 3rd stair ($|10-10| = 0$ energy lost).
So, the total energy lost is 0 which is the minimum.
Hence, the answer is 0.

Sample Input 2:

2

8

7 4 4 2 6 6 3 4

6

4 8 3 10 4 4

Sample Output 2:

Previous

Next

1

2

3

4

5

6

7

8

9

10

11

12

13

#include<bits/stdc++.h>

int f(int ind, vector<int> &heights) {

if(ind == 0) return 0;

int left = f(ind-1, heights) + abs(heights[ind] - heights[ind-1]);

int right = INT_MAX;

if(ind > 1) right = f(ind-2, heights) + abs(heights[ind] - heights[ind-2]);

return min(left, right);

}

int frogJump(int n, vector<int> &heights)

{

return f(n-1, heights);

}

Sample Test Case

Custom Test Case

Test Case 1

Correct Answer

Test Case 2

Correct Answer

Test Case 3

Correct Answer

Test Case 4

Time Limit Exceeded

Test Case 5

Time Limit Exceeded

Show Hint

Last saved on 6:43:25 PM

Run





C++ (g++ 5.4)

1 day ago Correct Answer 100 0%

Correct Answer

▶ Run

2.50

 $f(\text{nd})$

{

$\text{if } (\text{nd} == 0) \text{ return } 0;$

 $\text{left} = f(\text{nd} - 1) + \text{abs}(a[\text{nd}] - a[\text{nd} - 1]);$ $\text{right} = \text{INT} - \text{MAX};$ $\text{if } (\text{nd} > 1)$ $\text{right} = f(\text{nd} - 2) + \text{abs}(a[\text{nd}] - a[\text{nd} - 2]);$ $\text{return min}(\text{left}, \text{right});$

(1, 2, 3, 4, ..., n-1)



2.50

```
dp[0] = 0  
for (i = 1 → n-1)  
{  
    js = dp[ind-1] + abs(a[ind] - a[ind-1])  
    ss = dp[ind-2] + abs(a[ind] - a[ind-2]);  
    dp[ind] = min(js, ss);  
}
```



Problem

Submissions



Correct Answer

Report

Submitted on Jan 8, 2022, 6:55:02 PM

Penalty

Score ?

0%

100

Runtime

Language

277ms

C++ (g++ 5.4)

Previous Submissions

Submitted On	Status	Score	Penalty
10 mins ago	Correct Answer	100	0%
1 day ago	Correct Answer	100	0%
1 day ago	Correct Answer	100	0%
1 day ago	Correct Answer	100	0%

```
1 #include<bits/stdc++.h>
2 int frogJump(int n, vector<int> &heights)
3 {
4     vector<int> dp(n, 0);
5     dp[0] = 0;
6     for(int i = 1; i < n; i++) {
7         int fs = dp[i-1] + abs(heights[i] - heights[i-1]);
8         int ss = INT_MAX;
9         if(i > 1) ss = dp[i-2] + abs(heights[i] - heights[i-2]);
10
11         dp[i] = min(fs, ss);
12     }
13     return dp[n-1];
14 }
```

Sample Test Case

Custom Test Case

Test Case 1



Correct Answer

Test Case 2



Correct Answer

Test Case 3



Correct Answer

Test Case 4



Correct Answer

Test Case 5



Correct Answer



Previous



Next



Show Hint



Last saved on 6:55:09 PM



Run Code



30:57 / 38:49

Implementing tabulation >



2.50

$dp[i-2]$ $dp[i-1]$ $dp[i]$
 $dp[i-2]$ $dp[i-1]$ $dp[i]$
 0 1 2 3 4 ...

$prev2$ $prev$ cur
 $prev2$ $prev$ cur



$\{$
 $\underline{\underline{prev2}} = prev$
 $prev = cur$

2.50

(Top-Down)

Bottom Up

~~int dp[n]~~ → 0

~~dp[0] = 0~~

prev = 0
prev2 = 0

i = n

for (i = 1 → n-1)
{
 prev

js = ~~dp[i-1]~~ + abs(a[i] - a[i-1])

if (i > 1) prev2

ss = ~~dp[i-2]~~ + abs(a[i] - a[i-2])

return



35:37 / 38:49

Space optimization >



2.50

$$ss = \cancel{dp[ind-2]} + abs(a[ind] - a[ind-2]);$$

$$\begin{aligned} & \text{cur} \\ & \cancel{dp[ind]} = \min(jc, ss); \\ & \text{prev2} = \text{prev} \\ & \text{prev} = \text{cur} \end{aligned}$$

print(prev)



Problem

Submissions



Correct Answer

Report

Submitted on Jan 8, 2022, 7:00:59 PM

Penalty

Score ?

0%

100

Runtime

Language

256ms

C++ (g++ 5.4)

Previous Submissions

Submitted On	Status	Score	Penalty
5 mins ago	Correct Answer	100	0%
10 mins ago	Correct Answer	100	0%
1 day ago	Correct Answer	100	0%
1 day ago	Correct Answer	100	0%
1 day ago	Correct Answer	100	0%

```
1 #include<bits/stdc++.h>
2 int frogJump(int n, vector<int> &heights)
3 {
4     int prev = 0;
5     int prev2 = 0;
6     for(int i = 1; i < n; i++) {
7         int fs = prev + abs(heights[i] - heights[i-1]);
8         int ss = INT_MAX;
9         if(i > 1) ss = prev2 + abs(heights[i] - heights[i-2]);
10
11         int curi = min(fs, ss);
12         prev2 = prev;
13         prev = curi;
14     }
15     return prev;
16 }
```

Sample Test Case

Custom Test Case

Test Case 1



Correct Answer

Test Case 2



Correct Answer

Test Case 3



Correct Answer

Test Case 4



Correct Answer

Test Case 5



Correct Answer



Previous



Next



Show Hint



Last saved on 7:01:09 PM



Run Code



36:23 / 38:49

Space optimization >

